



“Python Programming”

Assignment-5(Capstone)

Topic – End-to-End Energy Consumption Analysis and
Visualisation

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Course – B. Tech CSE (AI & ML)

Section –B

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Introduction

This project focuses on analysing electricity usage across multiple campus buildings. By automating data loading, cleaning, aggregation, and visualisation, the system helps identify usage patterns, peak hours, and high-consumption buildings. The goal is to support better energy management and decision-making.

Objectives

- To read and validate building-wise energy CSV files automatically.
- To compute daily, weekly, and building-level energy statistics.
- To use object-oriented programming for structured data handling.
- To create a dashboard showing energy trends and comparisons.
- To generate summary files for administrative insights.

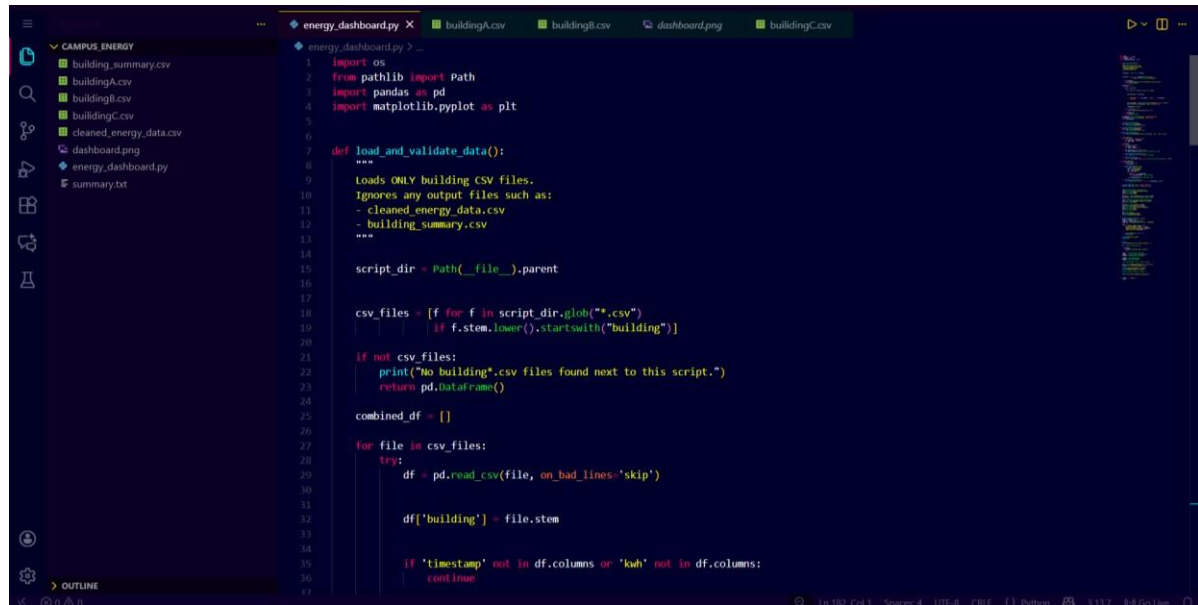
Program Description

The Python script loads all building CSV files from the same folder, cleans the data, and combines them into a single dataset. It calculates daily and weekly totals using Pandas and summarises consumption for each building.

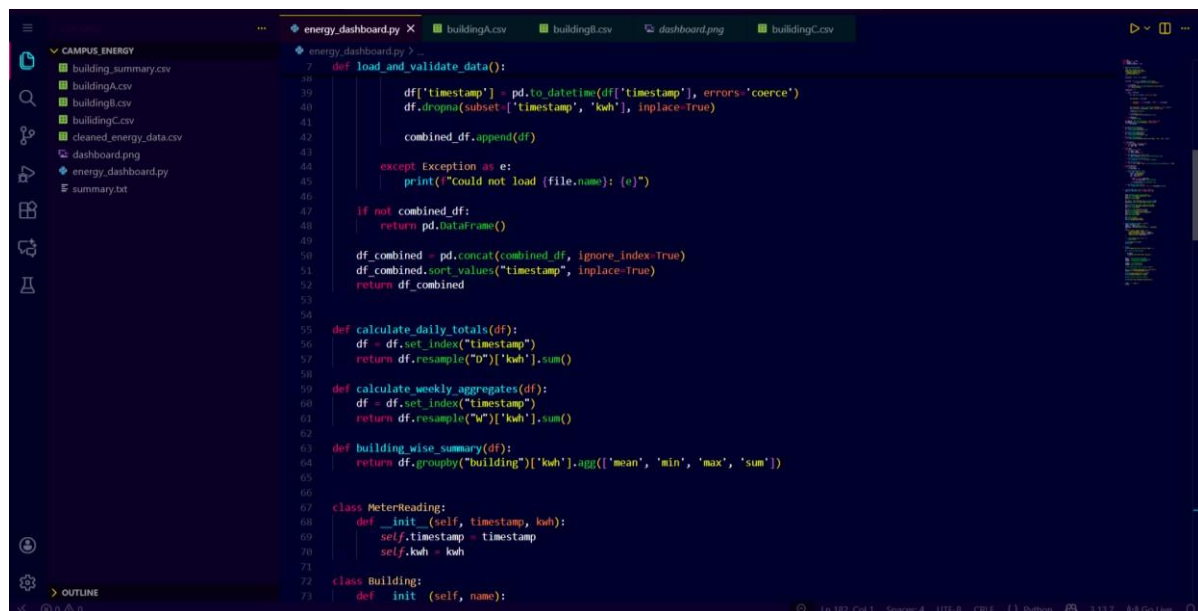
OOP classes are used to model buildings and meter readings. A dashboard is created using Matplotlib, containing a daily trend

line, weekly comparison bar chart, and hourly scatter plot. The program finally exports a cleaned dataset, building summary, and a text-based executive summary.

Program Code



```
1 import os
2 from pathlib import Path
3 import pandas as pd
4 import matplotlib.pyplot as plt
5
6
7 def load_and_validate_data():
8     """
9     Loads ONLY building CSV files.
10     Ignores any output files such as:
11     - cleaned_energy_data.csv
12     - building_summary.csv
13     """
14
15     script_dir = Path(__file__).parent
16
17     csv_files = [f for f in script_dir.glob("**.csv")
18                  if f.stem.lower().startswith("building")]
19
20     if not csv_files:
21         print("No building*.csv files found next to this script.")
22         return pd.DataFrame()
23
24     combined_df = []
25
26     for file in csv_files:
27         try:
28             df = pd.read_csv(file, on_bad_lines='skip')
29
30             df['building'] = file.stem
31
32             if 'timestamp' not in df.columns or 'kwh' not in df.columns:
33                 continue
34
35         except Exception as e:
36             print(f"Could not load {file.name}: {e}")
37
38     if not combined_df:
39         return pd.DataFrame()
40
41     df_combined = pd.concat(combined_df, ignore_index=True)
42     df_combined.sort_values("timestamp", inplace=True)
43     return df_combined
44
45
46 def calculate_daily_totals(df):
47     df = df.set_index("timestamp")
48     return df.resample("D")["kwh"].sum()
49
50
51 def calculate_weekly_aggregates(df):
52     df = df.set_index("timestamp")
53     return df.resample("W")["kwh"].sum()
54
55
56 def building_wise_summary(df):
57     return df.groupby("building")["kwh"].agg(['mean', 'min', 'max', 'sum'])
58
59
60 class MeterReading:
61     def __init__(self, timestamp, kwh):
62         self.timestamp = timestamp
63         self.kwh = kwh
64
65
66 class Building:
67     def __init__(self, name):
```



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campus_energy
energy_dashboard.py X buildingA.csv buildingB.csv dashboard.png buildingC.csv

CAMPUS ENERGY
building_summary.csv
buildingA.csv
buildingB.csv
buildingC.csv
cleaned_energy_data.csv
dashboard.png
energy_dashboard.py
summary.txt

class Building:
    def __init__(self, name):
        self.name = name
        self.meter_readings = []

    def add_reading(self, timestamp, kwh):
        self.meter_readings.append((timestamp, kwh))

    def calculate_total_consumption(self):
        return sum(r.kwh for r in self.meter_readings)

    def generate_report(self):
        return f"({self.name}): Total = {self.calculate_total_consumption():.2f} kwh"

class BuildingManager:
    def __init__(self):
        self.buildings = {}

    def ingest_dataframe(self, df):
        for _, row in df.iterrows():
            name = row['building']
            ts = row['timestamp']
            kwh = row['kwh']

            if name not in self.buildings:
                self.buildings[name] = Building(name)

            self.buildings[name].add_reading(ts, kwh)

    def generate_all_reports(self):
        return [b.generate_report() for b in self.buildings.values()]

def create_dashboard(df, daily, weekly, summary):
    fig, ax = plt.subplots(3, 1, figsize=(12, 16))
```

```
def create_dashboard(df, daily, weekly, summary):
    fig, ax = plt.subplots(3, 1, figsize=(12, 16))

    ax[0].plot(daily.index, daily.values)
    ax[0].set_title("Daily consumption Trend")
    ax[0].set_xlabel("Date")
    ax[0].set_ylabel("kwh")

    df['week'] = df['timestamp'].dt.isocalendar().week
    weekly_avg = df.groupby("building")["kwh"].mean()

    ax[1].bar(weekly_avg.index, weekly_avg.values)
    ax[1].set_title("Avg Weekly Usage per Building")
    ax[1].set_ylabel("kwh")

    df['hour'] = df['timestamp'].dt.hour
    ax[2].scatter(df['hour'], df['kwh'])
    ax[2].set_title("hourly Peak Consumption")
    ax[2].set_xlabel("hour")
    ax[2].set_ylabel("kwh")

    plt.tight_layout()
    plt.savefig("dashboard.png")
    print("✓ dashboard.png saved")

def generate_summary_report(df, summary):
    total = df['kwh'].sum()
    highest = summary['sum'].idxmax()
    peak_time = df.loc[df['kwh'].idxmax(), 'timestamp']

    text = (
        "=== ENERGY SUMMARY REPORT ===\n"
        f"Total consumption: {total:.2f} kwh\n"
        f"Highest Consuming Building: {highest}\n"
        f"Peak load time: {peak_time}\n"
    )

    with open("summary.txt", "w") as f:
        f.write(text)

    print("summary.txt saved")
    print(text)
```

```
def generate_summary_report(df, summary):
    text = (
        "=== ENERGY SUMMARY REPORT ===\n"
        f"Total consumption: {total:.2f} kwh\n"
        f"Highest Consuming Building: {highest}\n"
        f"Peak load time: {peak_time}\n"
    )

    with open("summary.txt", "w") as f:
        f.write(text)

    print("summary.txt saved")
    print(text)

def main():
    print("Loading CSV files from this folder...")

    df = load_and_validate_data()

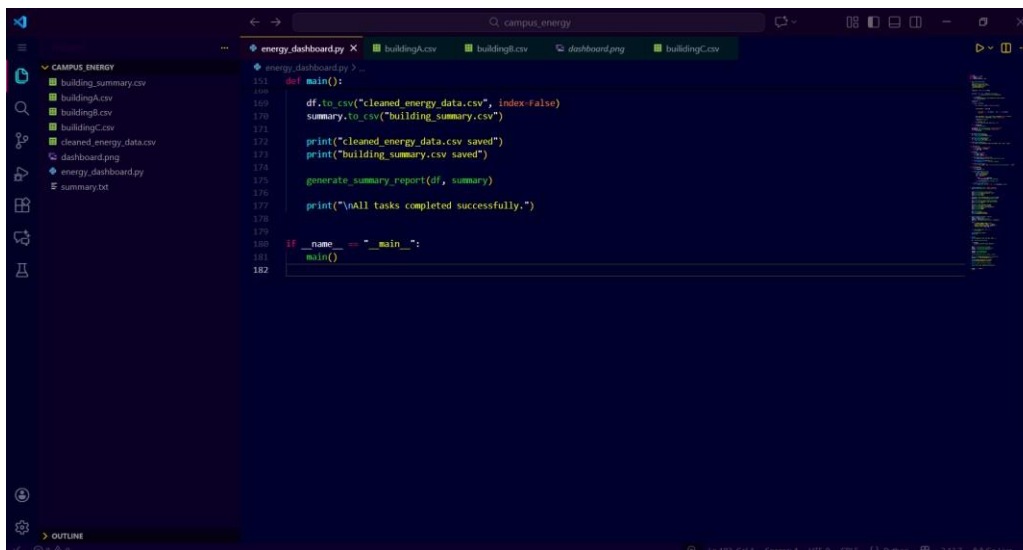
    if df.empty:
        print("No valid CSVs found. Exiting.")
        return

    daily = calculate_daily_totals(df)
    weekly = calculate_weekly_aggregates(df)
    summary = building_wise_summary(df)

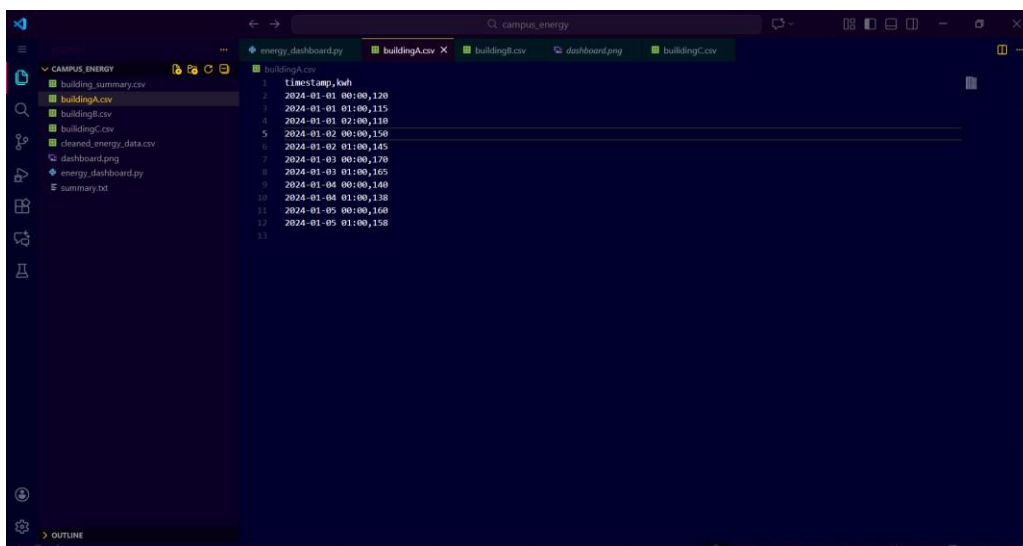
    manager = BuildingManager()
    manager.ingest_dataframe(df)

    create_dashboard(df, daily, weekly, summary)

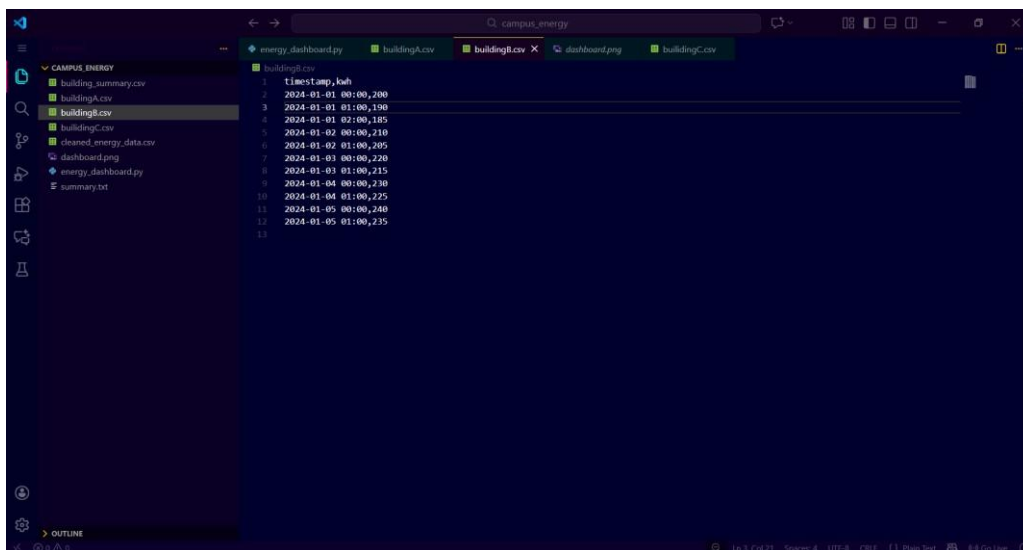
    df.to_csv("cleaned_energy_data.csv", index=False)
    summary.to_csv("building_summary.csv")
```



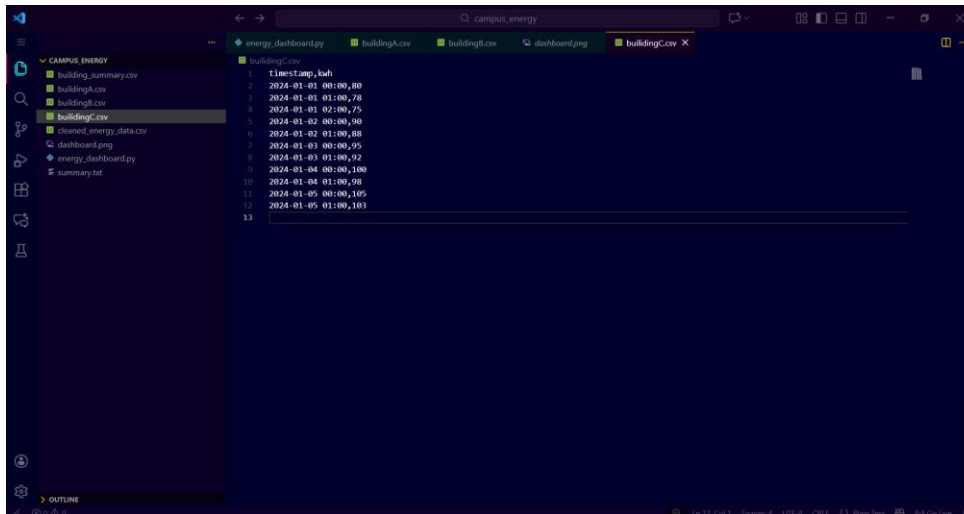
```
151 def main():
152     df.to_csv("cleaned_energy_data.csv", index=False)
153     summary.to_csv("building_summary.csv")
154
155     print("cleaned energy data.csv saved")
156     print("building summary.csv saved")
157
158     generate_summary_report(df, summary)
159
160     print("\nAll tasks completed successfully.")
161
162 if __name__ == "__main__":
163     main()
```



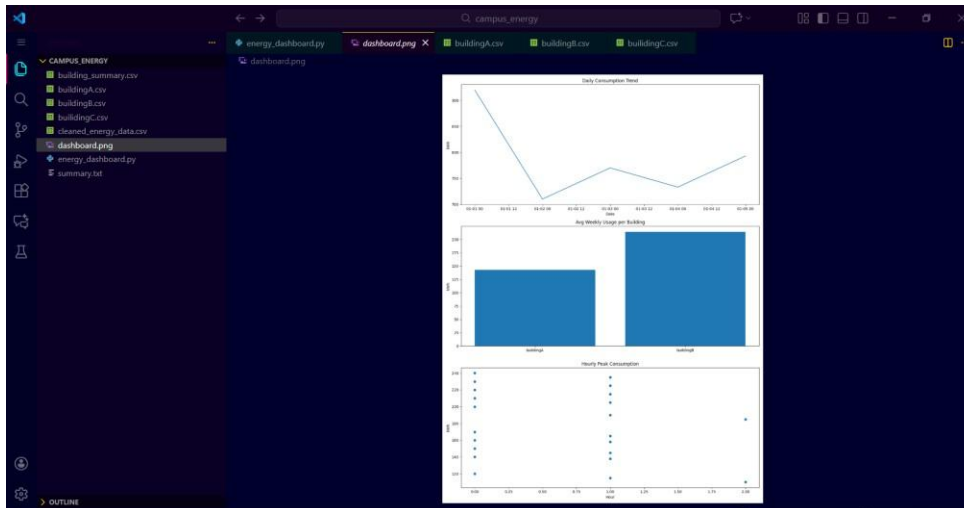
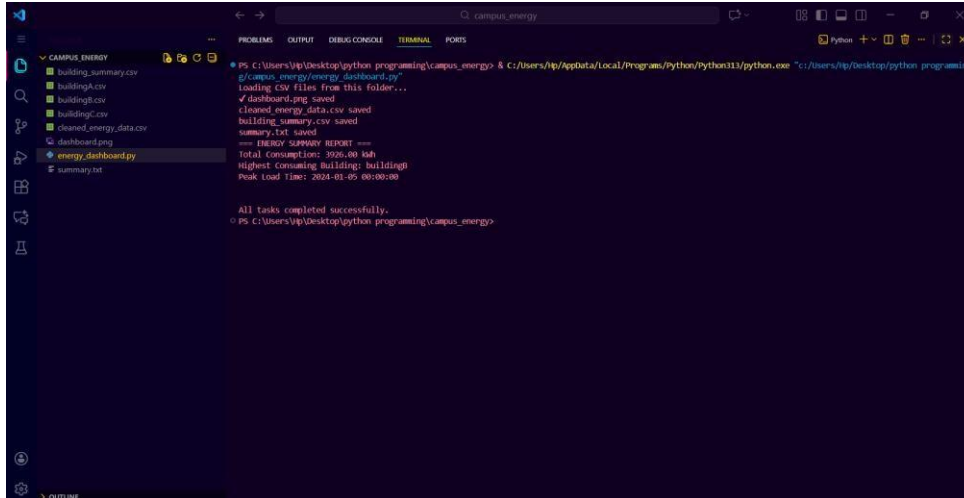
```
1 timestamp,kwh
2 2024-01-01 00:00,120
3 2024-01-01 01:00,115
4 2024-01-01 02:00,110
5 2024-01-02 00:00,150
6 2024-01-02 01:00,145
7 2024-01-03 00:00,170
8 2024-01-03 01:00,165
9 2024-01-04 00:00,140
10 2024-01-04 01:00,138
11 2024-01-05 00:00,160
12 2024-01-05 01:00,158
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```

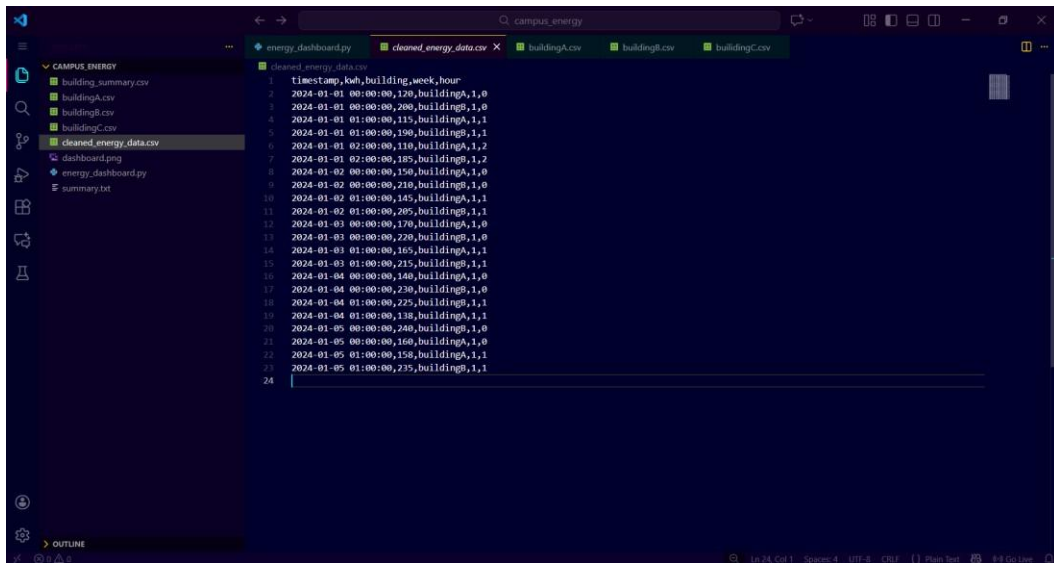


```
1 timestamp,kwh
2 2024-01-01 00:00,200
3 2024-01-01 01:00,190
4 2024-01-01 02:00,185
5 2024-01-02 00:00,210
6 2024-01-02 01:00,205
7 2024-01-03 00:00,220
8 2024-01-03 01:00,215
9 2024-01-04 00:00,230
10 2024-01-04 01:00,225
11 2024-01-05 00:00,240
12 2024-01-05 01:00,235
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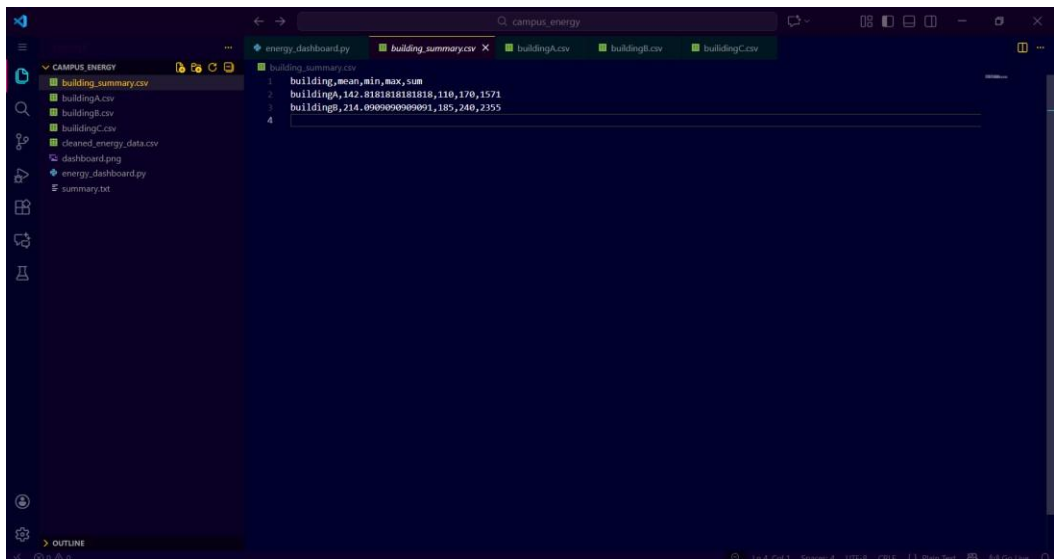


Sample Output

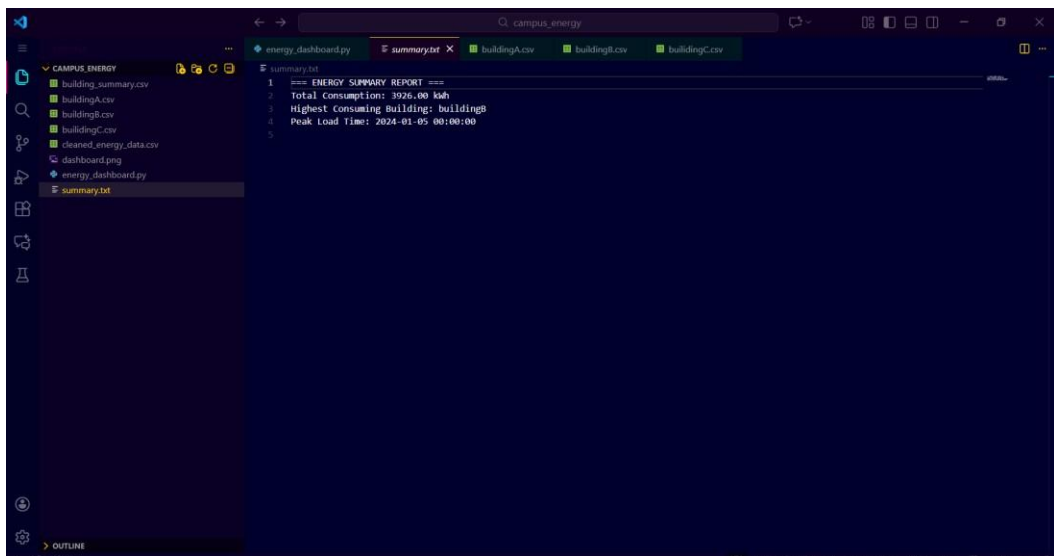




```
1 timestamp,kWh,building,week,hour
2 2024-01-01 00:00:00,120,buildingA,1,0
3 2024-01-01 00:00:00,200,buildingB,1,0
4 2024-01-01 01:00:00,115,buildingA,1,1
5 2024-01-01 01:00:00,150,buildingB,1,1
6 2024-01-01 02:00:00,110,buildingA,1,2
7 2024-01-01 02:00:00,185,buildingB,1,2
8 2024-01-02 00:00:00,150,buildingA,1,0
9 2024-01-02 00:00:00,210,buildingB,1,0
10 2024-01-02 01:00:00,165,buildingA,1,1
11 2024-01-02 01:00:00,205,buildingB,1,1
12 2024-01-03 00:00:00,170,buildingA,1,0
13 2024-01-03 00:00:00,220,buildingB,1,0
14 2024-01-03 01:00:00,165,buildingA,1,1
15 2024-01-03 01:00:00,215,buildingB,1,1
16 2024-01-04 00:00:00,140,buildingA,1,0
17 2024-01-04 00:00:00,230,buildingB,1,0
18 2024-01-04 01:00:00,225,buildingA,1,1
19 2024-01-04 01:00:00,138,buildingB,1,1
20 2024-01-05 00:00:00,240,buildingA,1,0
21 2024-01-05 00:00:00,160,buildingB,1,0
22 2024-01-05 01:00:00,158,buildingA,1,1
23 2024-01-05 01:00:00,235,buildingB,1,1
24
```



```
1 building,mean,min,max,sum
2 buildingA,142.8181818181818,110,170,1571
3 buildingB,214.0909090909091,185,240,2355
4
```



```
1 === ENERGY SUMMARY REPORT ===
2 Total Consumption: 3926.00 kWh
3 Highest Consuming Building: buildingB
4 Peak Load Time: 2024-01-05 00:00:00
5
```

Conclusion

The project successfully analyzes campus electricity usage and produces clear visual and textual insights. It automates data processing, identifies key consumption trends, and helps administrators make informed decisions for energy conservation.